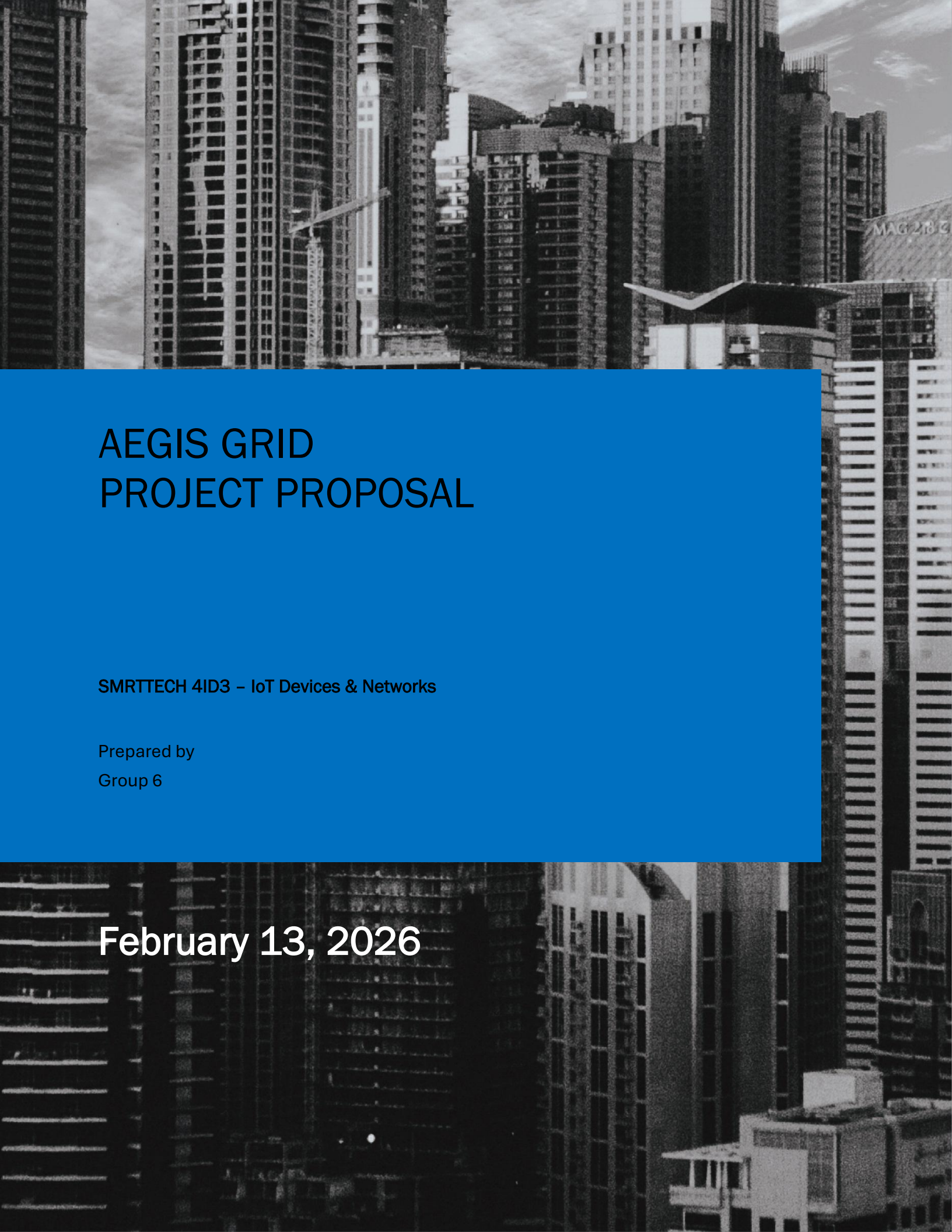

AEGIS GRID
PROJECT PROPOSAL

SMRTTECH 4ID3 – IoT Devices & Networks

Prepared by
Group 6

February 13, 2026

Aegis Grid
Project Proposal

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Nicolas Jao	400450151
Ayesha Mehmood	400403482
Yassin Merdan	400376619
Yihang Qin	400461329

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Introduction

Aegis Grid is a modern IoT solution purpose-built to solve one of the electrical grid's most persistent challenges: slow fault detection and limited real-time situational awareness. By deploying intelligent sensors across substations, feeder lines, and distribution infrastructure, Aegis Grid enables utilities to pinpoint outages with precision instead of relying on time-consuming field searches. When a fault occurs, distributed monitoring nodes instantly compare current flow across the network—allowing operators to isolate the exact section between substations within seconds and dramatically reduce restoration time. Beyond rapid fault localization, Aegis Grid delivers continuous, high-resolution visibility into grid health by monitoring load, voltage, temperature, and other early physical signs of equipment degradation. Advanced analytics and machine learning models transform this constant stream of data into predictive insights, detecting subtle anomalies before they escalate into failures and automatically alerting operators to emerging risks. Designed with scalability, resilience, and operator safety in mind, Aegis Grid empowers utilities to transition from reactive outage response to proactive infrastructure management, minimizing downtime, improving reliability, and strengthening the stability of the modern power system.

Stakeholders

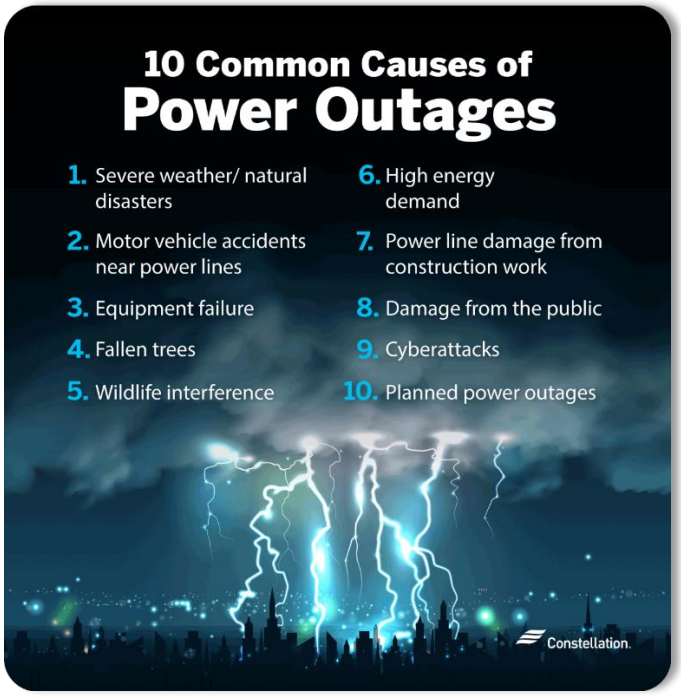
This is a list of all the parties that would be involved in this project or would have their livelihoods affected by this project.

Stakeholders	Goals/Objectives
Power Grid Companies	• Responsible for delivering reliable and safe electricity • Minimizing power outages caused by equipment failure • Provides real-time monitoring of voltage, current and temperature
Grid Operators	• Responsible for monitoring grid performance (making sure grids are operating properly) • Repair failed power grids
Municipalities	• Depends on grids to power infrastructure such as transportation and hospitals • Ensure public safety during power outages
Emergency Services (911 Operators, Ambulance etc.)	• Depends on grids for continuous electricity for critical operations/emergencies • Ensure public safety during power outages
Homeowners	• Depends on grids for essentials such as heating, lighting and cooling • Ensure comfortable and safe living conditions
Electricity Users (Businesses & Civilians)	• Depends on grids for daily operations • Minimize downtime when power outages occur

Government	• Sets standards and regulations to ensure power systems are safe for workers and the public

Problem Statement

Every year, power grids experience widespread failures and costly downtime due to aging infrastructure, equipment degradation, insulation breakdown, overheating components, and overloaded transmission assets. Many critical devices—such as transformers, circuit breakers, and generators—operate continuously under electrical and mechanical stress, gradually increasing the risk of unexpected faults and sudden outages. These failures are often detected only after performance has already deteriorated or service has been disrupted, as traditional monitoring approaches rely heavily on periodic inspections and reactive maintenance practices. As grid infrastructure continues to age and demand on electrical systems increases, conventional maintenance strategies are no longer sufficient to ensure reliable and safe delivery of power. The resulting disruptions affect critical services, industrial operations, and public safety, emphasizing the urgent need for real-time condition monitoring and predictive maintenance solutions to enhance grid resilience and operational reliability.



Aegis Grid will focus on causes 3 and 6.

Aegis Grid aims to mostly focus on the third most common cause of power grid failures: equipment failure. The other cause of power grid failure that Aegis Grid will protect against are high energy demand spikes, as it is related to the third cause.

How Will It Affect Stakeholders?

Beneficiaries / External Stakeholders

The primary beneficiaries of Aegis Grid are power utilities, businesses, and the general public who rely on stable and reliable electricity. By providing immediate localization of power grid failures, real-time monitoring of grid health, and predictive alerts for equipment failure, the system helps reduce the likelihood of outages, equipment failures, and large-scale power disruptions. Consumers benefit indirectly through improved continuity of service, fewer disruptions to daily life, and enhanced safety. Power companies gain operational efficiency by being able to find failed power grid lines fast and pre-emptively addressing potential failures, reducing emergency maintenance costs and minimizing downtime. Regulators and policymakers also benefit as the technology supports compliance with grid reliability standards and encourages proactive infrastructure management.

Aegis Grid provides safety for civilians and economic recovery for businesses.

Project Creators / Internal Stakeholders

The team developing the project, including engineers, designers, and researchers, gains hands-on experience with IoT integration, sensor networks, and predictive monitoring systems. This enhances technical skills, fosters innovation, and contributes to professional development.

The project also encourages collaboration, problem-solving, and project management skills, as team members must coordinate hardware design, software programming, and data communication protocols. For educational or startup contexts, successfully demonstrating the concept can lead to recognition, potential funding opportunities, and credibility in the energy technology and IoT sectors. Aegis Grid presents an opportunity to build the modern IoT industry in the power sector, leading to the widely dreamt smarter cities of the future!

Indirect / Long-Term Impacts

In the long-term, the successful implementation and scaling of such a system could influence wider infrastructure resilience strategies and set standards for smart-grid monitoring solutions. It may encourage adoption of similar IoT-based predictive maintenance technologies in other critical infrastructure sectors, benefiting society at large by increasing safety, reliability, and efficiency. Aegis Grid would be the catalyst for this bright future!

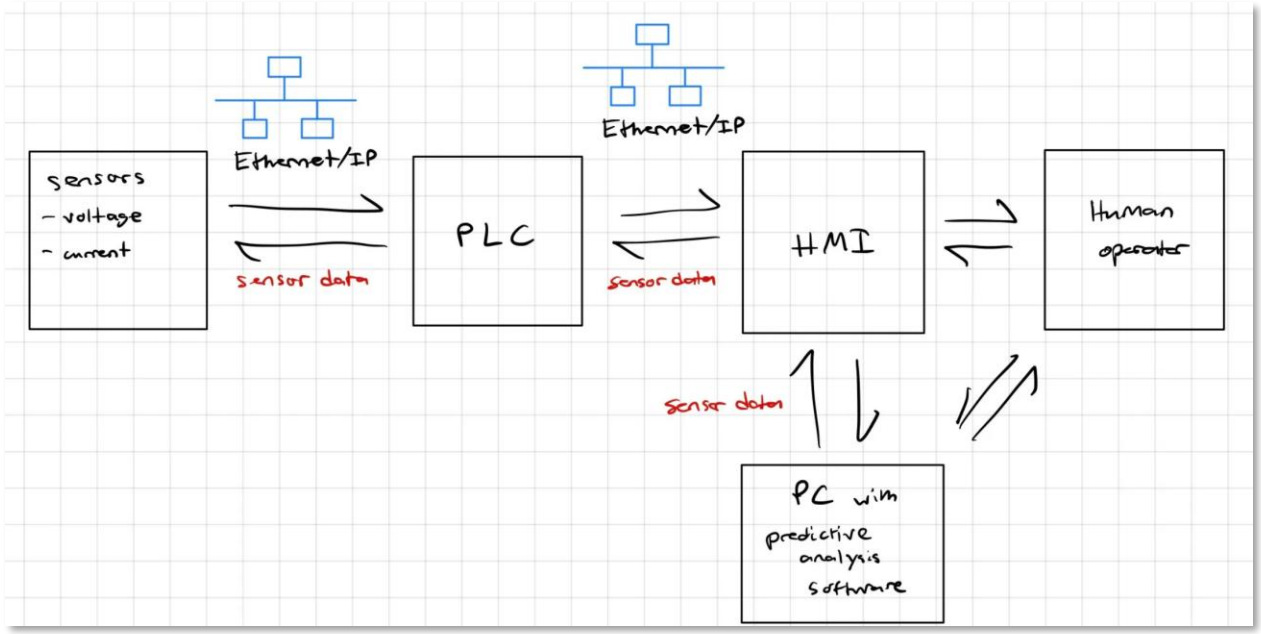
What Is The Current Solution?

Today, power grids are mainly monitored using centralized control systems such as SCADA, along with a limited number of sensors installed at substations and key nodes. These legacy systems are still necessary—they collect enormous amounts of data from power grids and visualize it to operators for maintenance. The problem is collecting the data itself—many power grids today still do not have an efficient way to collect data from the equipment in the field, or their current sensor deployment is subpar and methods of collecting data can be vastly improved. As a result, grid operators also rely on scheduled inspections and maintenance reports to assess grid conditions and respond to potential issues (Safegrid, 2023). While these methods provide a general overview of grid performance, they do not offer detailed or continuous insight into conditions along transmission and distribution lines, where there is traditionally a lack of data collection. In many cases, faults caused by equipment degradation or sudden demand spikes are only identified once a failure has already occurred (Tsioumpri, 2021).

Issues with current power grid monitoring:

- Limited real-time visibility. Traditional SCADA and legacy systems struggle with high-speed data and predictive monitoring. Outage time can also be drastically reduced with more data collection as utilities can pinpoint the exact location of the failure instead of the general feeder line it had occurred
- Manual inspections and slow fault detection. Many assets are at remote locations and are hard to monitor continuously, and sending utility teams can be costly and time-consuming
- Low efficiency and delayed decision-making due to lack of detailed real-time data. Decisions are slower because there is not enough information to make them well. Information is centralized at a few substations and the main facilities
- Complex, aging infrastructure and poor interoperability between old and new systems. Aegis Grid aims to focus on data collection, then the data will be transformed and communicated efficiently for the legacy SCADA systems

This is an example of a network setup for current solutions in the industry:

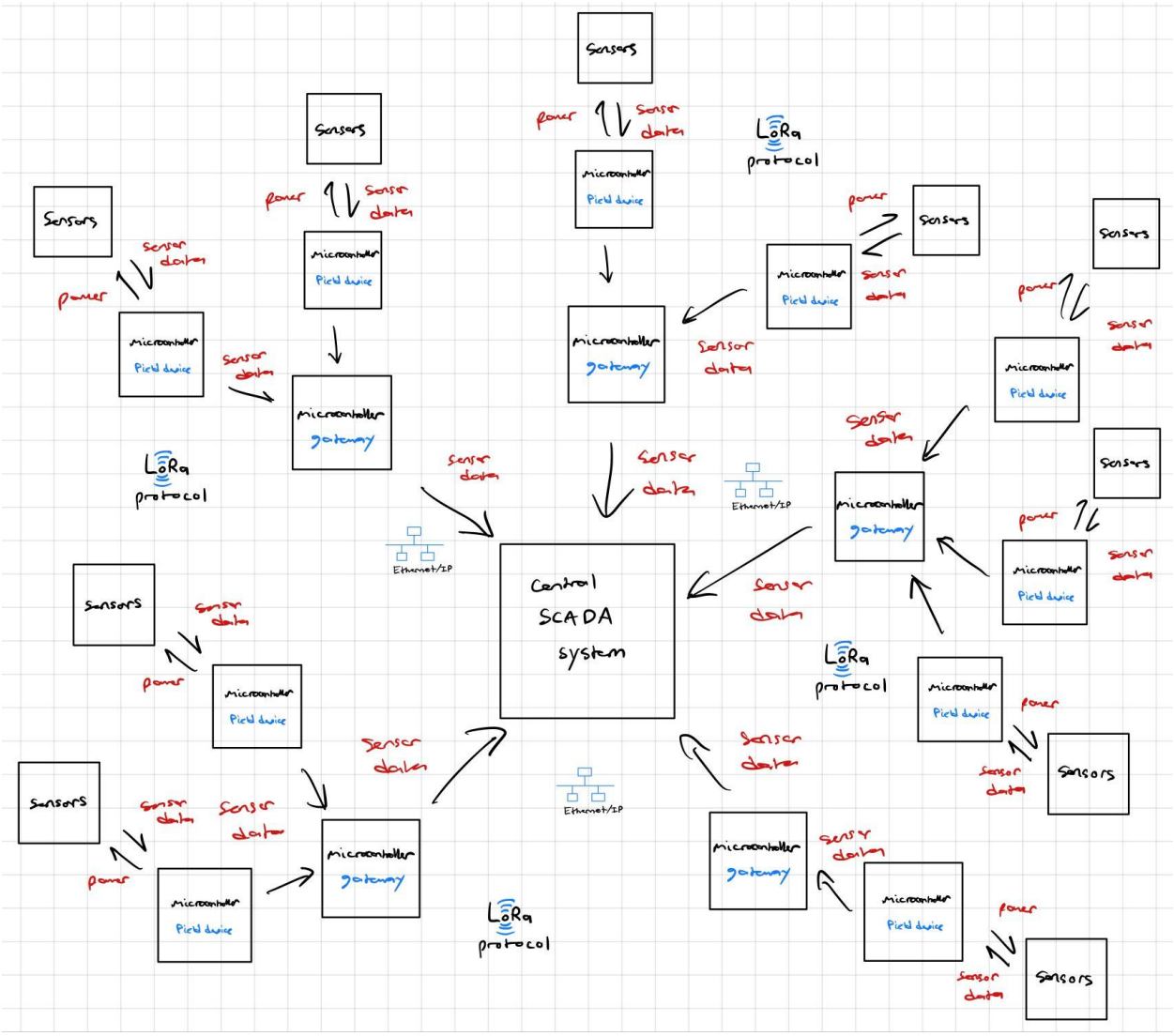


How Will *Aegis Grid* Be Better?

Aegis Grid improves upon current solutions by shifting grid monitoring from a reactive approach to a preventive and data-driven one, with the main focus on detecting early signs of equipment failure through sensors in the field and a robust IoT network. Instead of relying on a small number of centralized sensors, periodic manual inspections, and after-the-fact fault reports—which often identify problems only after damage has occurred—Aegis Grid deploys distributed sensors directly on transformers, conductors, breakers, and other critical assets across the entire power grid. This allows grid operators to observe equipment health in real time, including abnormal vibrations, insulation degradation, overheating, load imbalances, and electrical anomalies that signal impending mechanical or electrical failure.

The main value proposition lies in improving the abundance of data collection for faster and more informed decision-making, saving time, current costs, and future costs for stakeholders.

This is a proposed possibility for an IoT network setup for the industrial application of Aegis Grid:



What is better about this network than the previous one?

- The real-time monitoring range is increased. With inexpensive sensors, gateways, and field devices, sensors can be deployed everywhere for a much larger capability of data collection. Remote locations of the power grid can now be reached
- The system is scalable. Easily add more sensors or monitor multiple assets without heavy infrastructure changes
- An existing IoT network architecture is added upon the legacy systems. This addition of technology, rather than replacement, ensures quick implementation and compatibility with the crucial SCADA and PLC legacy systems that current power grid facilities depend on
- Secure and reliable. LoRa offers encrypted, bidirectional communication for safe, consistent monitoring

By collecting continuous on-site data, Aegis Grid can identify abnormal conditions before they develop into full system failures. Early warnings enable operators to take targeted action, such as load redistribution or preventive maintenance, reducing both outage frequency and downtime.

The IoT network also serves a second purpose: even after a breakout occurs, a current problem in the industry is that utilities will know which feeder line is down, but not the exact location of the problem. Sending a team to drive and find the site sometimes takes hours and prolongs the outage. Aegis Grid sensors will be distributed along each feeder line so that they provide real-time feedback between substations of power grid health, automatically signaling the site with the issue.

These are the sensors that will be deployed inside equipment all over the power grid for the industrial use application of Aegis Grid:

Root Cause	Early Physical Signal	Sensor
Insulation breakdown	partial discharge, gas formation	UHF antennas, capacitive couplers, bushing PD sensors, hydrogen/acetylene/methane sensors
Overload / Overheating	overcurrent, temperature rise, cooling failure	RTDs / thermocouples, ampmeters
Mechanical wear	vibration, acoustic emissions	accelerometers, acoustic emission sensors, vibration sensors
Electrical instability	voltage/current anomalies	voltage/current sensors, power quality analyzers

For an industrial application, these sensors must be numerous and deployed virtually everywhere along the power lines of entire towns and cities for Aegis Grid to be effective. Therefore, a large IoT network must be designed with robust communication protocols, and all sensors must be reliable, durable, long-lasting, and power-efficient in the field. Each one must also be connected to the cloud.

Because they are spread out everywhere, it allows for efficient preventative maintenance because when sensors in an area begin detecting rising abnormal conditions, a dispatch team from the power grid company could be deployed to investigate the area before things get critical. Critical conditions would be defined according to the conditions of the technology of the power grid, which is specified and tested by the manufacturers of the equipment and devices.

The system is scalable by simply adding additional sensors in new areas and connecting them to the cloud. Sensor data from electrical monitoring is transmitted over a simple communication network. We are choosing LoRa as the communication protocol for this application because it combines long-range connectivity with low power consumption, making it ideal for distributed power-grid sensors that may be installed in remote or difficult-to-access locations. Its ability to transmit small packets of data over several kilometers without relying on existing cellular infrastructure ensures that temperature, vibration, and current measurements from each sensor node can reliably reach a central monitoring hub. Additionally, LoRa supports secure, scalable networks with bi-directional communication, enabling real-time alerts and

configuration updates, which is essential for predictive maintenance. This balance of range, energy efficiency, and network robustness makes LoRa a practical choice for a smart grid IoT system that prioritizes continuous monitoring and early fault detection.

Overall, Aegis Grid offers better reliability, faster response times, and improved safety compared to existing systems, while remaining scalable and adaptable to future grid expansions.

Proof-of-Concept Solution

To demonstrate the core functionality of Aegis Grid, a small-scale proof-of-concept will be developed using low-voltage electronics and hobbyist-grade sensors to safely simulate real-world grid monitoring and protection scenarios. The core concept is to develop a small-scale replica of a piece of equipment in the power grid, inducing abnormal or low-health conditions, then testing if the sensors give the correct feedback.

For our objectives of general electrical safety and quick outage response times, miniature parallel feeder lines will be constructed using resistors on a breadboard. Sensors (**voltage comparator ICs**) will be placed along the lines and will send signals to the microcontroller when the voltages are lower or higher than our chosen warning threshold. This warning threshold will be created through circuit design and programming. When the microcontroller receives an input a voltage is outside of our warning threshold, meaning measured values deviated from predefined operating ranges, visual and audible indicators such as **LEDs** and **buzzers** will be activated as an alarm. Elevated current or voltage levels are used to simulate periods of high demand on the power grid, while abnormally low values represent aging infrastructure or equipment degradation.

For one feeder line, we will design a circuit with a series of nodes connected by **shunt resistors** to simulate line resistance and voltage drop. For example, let's say S is for "substation": S1 → 100Ω → S2 → 100Ω → S3 → 100Ω → S4. We will power the line with a DC voltage source (such as 5V from the microcontroller), and each substation is a voltage measurement point.

1. Line segments: S1 → S2 → S3 → S4
2. Place series resistors (simulate line resistance) between nodes
3. Insert a shunt resistor in series with each segment
4. Connect microcontroller analog pins across shunt resistors
5. Measure voltage → calculate current → determine which segment has zero current

We will simulate an outage situation by applying an open circuit at one of the nodes. Even if the entire feeder goes off, the last segment that still had current before the break tells us where the outage happened. We can also design the circuit to provide an input to the microcontroller to get the data and convert it to the substation string as an output to the serial monitor. Because

shunt resistors are not that cheap and we only need one to test this proof-of-concept idea, we may purchase a limited number of them.

To address the primary problem of detecting signs of equipment degradation, we will have multiple sensors on a circuit to simulate the detection of common signs of degradation as mentioned in the previous section, but with smaller-scale sensors.

These include:

- **Humidity sensors** (**analog**) to simulate detecting insulation breakdown. Many power grid components, like transformers, switchgears, and insulators, rely on dry environments. High humidity can penetrate insulation materials, causing them to absorb moisture, which can lead to partial discharges or short circuits
- **Vibration sensors** (**analog**) to simulate mechanical wear. Power grid equipment naturally produces vibration, but these vibrations are regular and stable when the equipment is healthy. As components degrade, vibrations change in frequency, amplitude, or pattern
- **Temperature sensors** (**analog**) to simulate overheating. A leading cause of equipment degradation is constant overheating, with too much heat causing damage to the equipment

A microcontroller with LoRa capabilities will control these sensors. The sensors will be attached with wires to the microcontroller (1-wire or 2-wire protocol). We will define the threshold values ourselves (they do not matter as long as we accomplish testing the proof-of-concept) and program each sensor accordingly. Then we will use the LEDs and buzzers to simulate the alerting system when the values go beyond the thresholds.

We will also use MQTT for a data visualization dashboard and network reliability. For the real industrial application, this network would be more advanced with multiple nodes.

Finally, we will need a device to test these sensors on as our “power grid equipment.” We plan to use a simple DC motor. This could be a generator or general load on the industrial application.

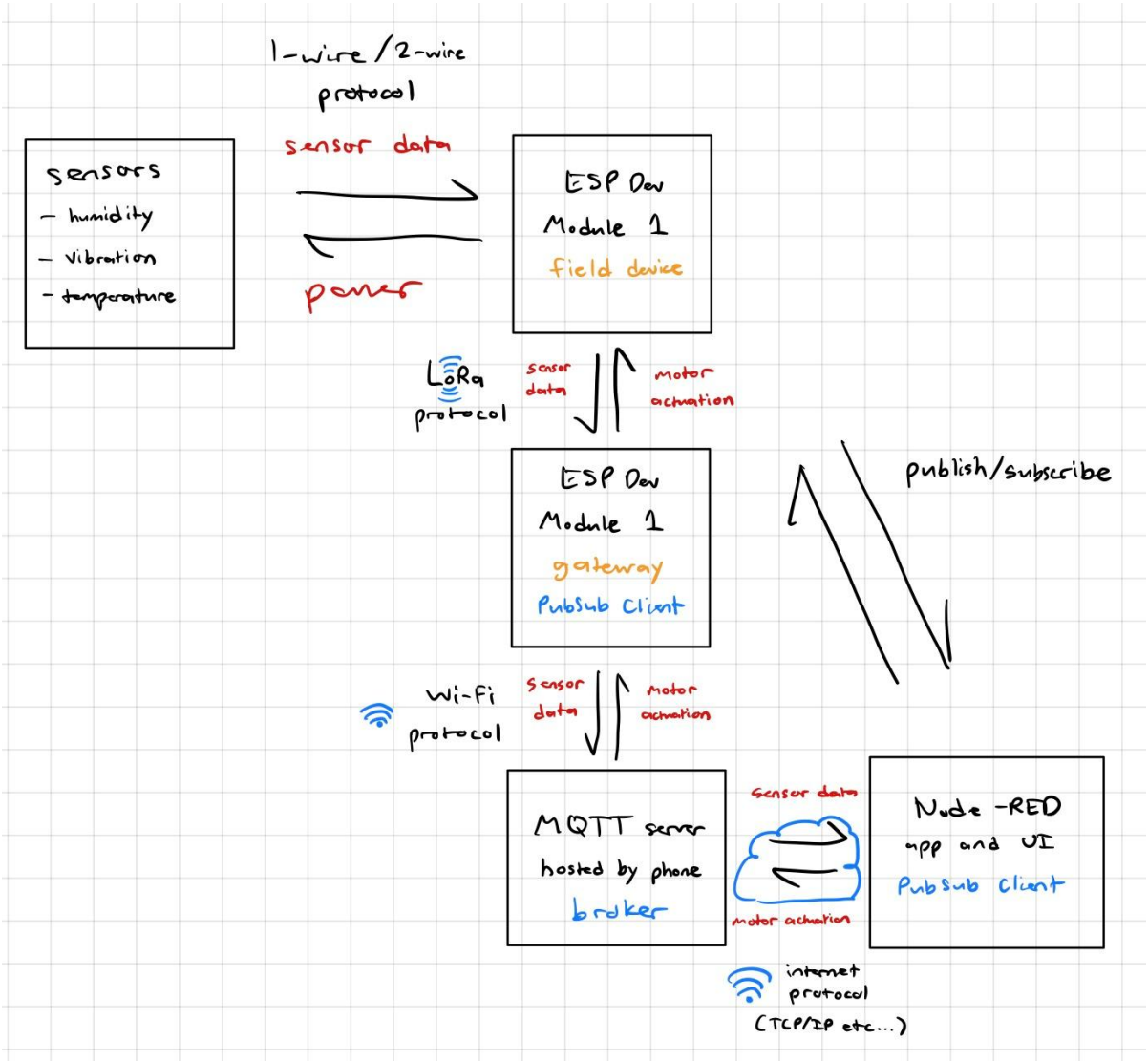
These are the conditions we will simulate to test each sensor:

- **Humidity:** Our small DC motor will not naturally create humidity. We must create an enclosed humidity chamber or a cup of water and pretend it is an environment that induces humid conditions for our motor that cause insulation breakdown
- **Vibration:** We will attach an irregular load to the shaft of the motor to induce irregular motion. We could tape an off-center mass such as an eraser to create an oscillation for the vibration sensor
- **Temperature:** We will intentionally increase the current supplied to the motor. We must run the motor and define a baseline operation temperature. Then we must deviate from that temperature. We could either do this by gently stalling the shaft with friction to increase its mechanical load and current draw or by increasing the voltage supply, whichever one is safer. Both would increase heat. To simulate cooling failure (when power equipment overheats because cooling systems fail), we could wrap the motor loosely with a cloth, put it in a box, and block airflow to cause heat to build up faster

For our IoT network architecture, we will have a layered star topology much like the industrial application solution. We chose this architecture because our goal is to deploy as many sensors as possible to equipment all over the power grid, and the power grid is all over the country with many remote locations. We will have multiple gateways, and then field devices that power the sensors. However, in our proof-of-concept, we will just have one gateway and one field device. This topology ensures efficient reach and data collection, as well as a quick way to localize network failures.

Gateway and field devices will be powered by batteries. With LoRa, these devices will last a long time, and with handshake communication from the protocol, maintenance is easy as issues can be localized quickly for repair. An MQTT server and a Node-RED dashboard will simulate the data being collected by the centralized SCADA system and visualized for operators. This is only for the proof-of-concept, as the communication would be more complex for the industrial application, as an integration plan between the IoT network and the legacy systems must be created and implemented.

This is a block diagram of our proof-of-concept IoT network:



To summarize, our plan includes building a circuit with a feeder line and voltage comparators to monitor irregular electricity demand. One feeder line will contain a shunt resistor so that we can test Aegis's Grid's goal of quick issue localization after an outage. We will have a DC motor and then artificially induce faulty conditions on it to test if our sensors will recognize the faults and interpret the data correctly for the LEDs and buzzer alarm.

This architecture illustrates how Aegis Grid integrates distributed sensing, real-time communication, and automated response into a unified, scalable system. It is a prime example of an IoT network solution for a modern-day problem.

Budget

This section covers the budget only for the proof-of-concept solution.

BOM

Qty	Description	Purchase link	Price
2	MacIoT board	register usage	$0.00
10	Low offset voltage dual comparators	Link	$11.50
1	DHT11 humidity sensor	we have	$14.00
1	Piezoelectric ceramic vibration sensor module	Link	$7.00
1	TMP36 temperature sensor	Link	$6.00
2	Shunt resistor	Link	$15.00
2	Red LED	we have	$0.50
1	active or passive buzzer	we have	$2
1	DC motor	we have	$14
1	breadboard	we have	$10
		Total:	$80

Estimated Total Cost of Project

~$80 CAD

This number includes the materials we already have and is an expected approximation.

Risks

In any innovative technical project, especially one involving emerging IoT technologies and integrated hardware, a comprehensive risk analysis is essential to ensure successful execution and reliable outcomes. This section outlines the key uncertainties that could affect the project's objectives, evaluates their potential impacts, and describes planned mitigation strategies to manage those risks effectively.

- Cyberattacks
  - Aegis Grid is specialized for equipment failure and high electricity demand. It does not cover cybersecurity vulnerabilities in the system of the power grid
  - Mitigation is not applicable as this is not part of the scope of Aegis Grid (perhaps a future expansion? Aegis Grid v2.0?)
- Prototype electrical hazards
  - The plan for the proof-of-concept prototype includes intentionally spiking voltages and currents to simulate high energy demand. This could potentially be dangerous
  - The proposed mitigation plan for this risk would be to use safer methods at creating the conditions for our motor for our sensors to detect
- Communication reliability
  - Wireless protocols may suffer interference, disconnections, or packet loss. With our chosen microcontroller's simple networking stacks, this is especially common. There is also the possibility of IoT network services such as MQTTool or Node-RED having unreliable communication (due to free versions or general website issues)
  - We will do our best to mitigate this by choosing stable communication protocols and retry logic and status conditions in our microcontroller code. This will be part of Aegis Grid's design of redundancy and reliability
- Cost vs. value
  - There is a chance our project's scope can creep or that even the industrial application of our prototype has costs that are too high for stakeholders to find any value in

- This risk could be mitigated if, when prepared for industrial applications, a cost analysis is done to determine if the money saved from this product is greater than the costs to implement it. If the cost analysis deems that it is not, then the project can be redesigned for cheaper materials or overhead reduction

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